

1.6: Absolute Value Functions

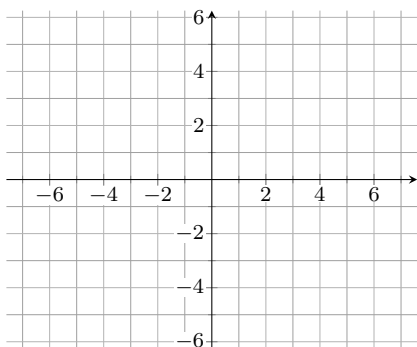
Definition. (Absolute Value Function)

The absolute value function can be defined as a piecewise function

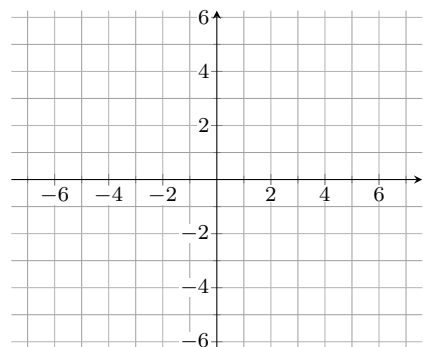
$$f(x) = |x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

Example. Graph the following

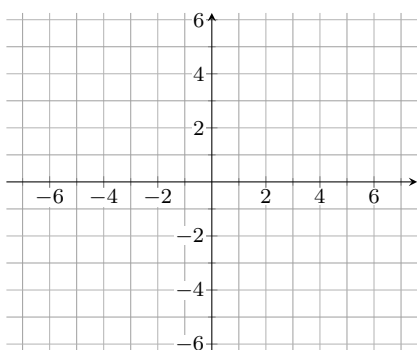
$$f(x) = |x|$$



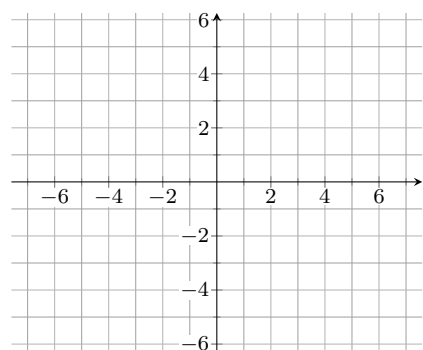
$$f(x) = |x - 2|$$



$$f(x) = |x| + 3$$



$$f(x) = 2|x - 3| + 1$$



Example. Solve the following equations

[Graphs](#)

$$|x| = 3$$

$$|2x + 6| = 2$$

$$|x + 3| - 1 = 0$$

$$2 - |4 - x| = 0$$

Example. Solve the following inequalities

[Graphs](#)

$$|x - 5| \leq 4$$

$$\left| \frac{x}{4} - 1 \right| < 2$$

$$|2 - x| \leq 3$$

$$|x - 3| > 2$$